

Model Failure Challenge Classroom Run Kit

TOK Cognitive Science Activity Bank • Natural Sciences • 25-35 min

Category	Natural Sciences	Time	25-35 min
Difficulty	Medium	ID	model-failure-challenge

Big idea: Models simplify reality productively.

One-Minute Teacher Brief

Students learn that models are useful simplifications rather than perfect copies of reality.

Student Prompt

Inspect Model Failure Challenge Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Model Failure Challenge Stimulus A	Student-facing stimulus 1: A simple weather, disease, or traffic model that works until one assumption changes. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Model Failure Challenge Stimulus B	Student-facing stimulus 2: An anomaly card that forces students to revise a model. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Model Failure Challenge Reveal / Twist	Student-facing stimulus 3: A model limits table: useful for, weak when, assumptions, and missing variables. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Models simplify reality productively.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Model Failure Challenge Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Model Failure Challenge Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Are models useful because they are false?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how models simplify reality productively.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Are models useful because they are false?

Setup Checklist

- Prepare a starter stimulus using: A simple weather, disease, or traffic model that works until one assumption changes.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- A simple weather, disease, or traffic model that works until one assumption changes.
- An anomaly card that forces students to revise a model.
- A model limits table: useful for, weak when, assumptions, and missing variables.

Resources To Use

- Resource pack: <resources/model-failure-challenge/index.html>
- Card deck CSV: <resources/model-failure-challenge/cards.csv>
- Visual prompt: <resources/model-failure-challenge/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Groups identify variables missing from the model: prior knowledge, sleep, stress, revision quality, teaching, test difficulty.
2. They draw a revised model with extra variables.
3. They decide whether the original model is useless, useful or dangerous.
4. Discuss when a model should be rejected.

Debrief Moves

- Knowledge question: Are models useful because they are false?

- How much simplification is acceptable?
- What is the difference between a model, a law and an explanation?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...